Hourly Temperature Forecasting Using Time-Series Methods

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Program: MSC in IT for Business Data Analytics

Module: Advanced Forecasting Techniques for Data Scientists

Academic Year/Semester: Fall 2025

Word Count[1]**:** 2807

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[1]Excluding the reports' cover page,table of contents.

Table of Contents

1.Introduction

From planning energy use to guiding farm decisions, from scheduling outdoor operations to assessing climate risks, there is always a need for accurate temperature forecasting. Hourly temperature prediction is meaningful and challenging for testing different methods on time series tasks due to the dynamic and nonlinear features of the weather pattern in general, and especially at the hourly scale of data. Recently, the growth of high-frequency weather measurement has let researchers look beyond immediate twists and turns but also push for more precise multi-day forecasts. This project aims to predict the hourly average temperature for 1–7 days ahead using a publicly available high-frequency weather dataset, measured every 10 minutes. The work blends classic statistical approaches with modern deep learning models to compare how they stack up. It is a unified test bed to explore strengths and limits in baseline, traditional, and advanced forecasting methods under one evaluation framework.

Besides achieving correct forecast accuracy, another aspiration of the project is to understand the structure of time series, data of temperatures, including daily periodicity, shorter term changes, and finally, the effect of longer-term dependencies. Hence, by exploring various models and following the same setup, the importance of each respective technique is illustrated, which focuses on dealing with seasonality, trend, and randomness present within high-frequency atmospheric observations. Finally, besides forecast accuracy, the importance of efficiency, explainability, and robustness, depending on various forecast horizons, is also worked towards, which is very important when it comes to real applications, where the selection of the right predictive technique to forecast temperatures needs to fulfill various practical constraints.

2.Project Definition & Forecasting Task

The aim of this project is to predict the average temperatures on an hourly basis by using time series models on high-frequency meteorological data. The dependent variable is calculated by summarizing the raw 10-minute data on temperatures into hourly temperatures, creating a clean and consistent time series suitable for forecasting. Although other meteorological variables are provided in the data, such as humidity, pressure, and wind speed, the scope is restricted to a univariate predictive approach. The univariate setup enables a comparison between various predictive techniques on an equal basis, not having to contend with the complications arising from secondary input variables.

The forecast procedure has a forecasting time span ranging from 1 to 7 days (from 24 to 168 hours), which allows examining the performance of the models on a number of predictive horizons, starting from the short to the medium ones. These horizons are important and applicable under several real-life conditions. Short term forecasts could apply to energy control, adjusting HVAC systems, and operational decisions, whereas medium predictive horizons could apply to farm planning, external scheduling, and finally to weather-related risk management.

By framing the task in such a manner, it creates a common and identical foundation regarding the assessment of various forecasting methods. In turn, it enables the project to assess the basic, classical statistical, and deep learning approaches under identical experimentation circumstances, which ultimately focuses on the difference between their performances with varying horizons. Organizing the task on a univariate time series along with various horizons enables it to focus on the basic time-series attributes of the temperatures and assess which methods are able to capture these attributes effectively.

3. Data Exploration & Preparation

3.1 Dataset Description

The data features high-resolution meteorological observations that are recorded every 10-minute interval for a period of about a year, from 2020-01-01 00:10:00 to 2021-01-01 00:00:00. This provides a total of 52,696 instances for 21 features. These features are particularly interesting in that they include air temperature (T), dew point temperature (Tdew), relative humidity (rh), atmospheric pressure (p), wind speed (wv, max. wv), precipitation features (rain, raining), and solar radiation features (SWDR, PAR). In relation to the forecasting research, this analysis aims to solve a univariate problem concerning hourly averaged air temperature (y), which can be obtained by aggregating the 10-minute values for air temperature to provide forecasting durations of 1-7 days, corresponding to 24-168 hours.

3.2 Data Quality Assessment

Preliminary analysis suggests a continuous data set with physically feasible values for temperatures from −3°C to 28°C, excluding any obvious sensor error conditions. Timestamps are properly interpreted, with no apparent repetition. Missing data for a 10-minute resolution are tacitly accounted for in this analysis aggregated over an hourly average, assuming that sufficient data are available for a particular hour. There are no actions taken to eliminate data points that are potential outliers, as the data are physically feasible for this area over this period.

3.3 Hourly Aggregation Process

The key preprocessing step involves processing 10-minute observations of temperature data into a detailed hourly series that can be used for forecasting.

Procedure:

1. Parse date column as datetime and set as index
2. Resample to hourly frequency using mean aggregation on temperature (T)
3. Retain only timestamp and aggregated temperature (y)

Result: 8,785 hourly observations forming a continuous series:

date	**y**
2020-01-01 00:00:00	0.578
2020-01-01 01:00:00	0.142
2020-01-01 02:00:00	-0.153
2020-01-01 03:00:00	-0.615
2020-01-01 04:00:00	-1.417

This step reduces high-frequency noise in the data while maintaining important patterns for a longer forecasting period.

3.4 Feature Engineering

To incorporate both classical statistical models and neural networks, additional temporal features related to calendars are incorporated into the hourly series.

Calendar Features:

- hour (0-23): Hour of day
- dayofweek (0-6): Day of week
- month (1-12): Month of year
- quarter (1-4): Quarter of year
- is_weekend (0/1): Binary weekend indicator

Cyclical Encodings (to avoid artificial discontinuities):

$$\text{hour\_sin} = \sin(2\pi \times \text{hour}/24), \quad \text{hour\_cos} = \cos(2\pi \times \text{hour}/24)$$

$$\text{dow\_sin} = \sin(2\pi \times \text{dayofweek}/7), \text{dow\_cos} = \cos(2\pi \times \text{dayofweek}/7)$$

3.5 Final Modeling Dataset

The complete, model-ready dataset contains:

Column	Type	Purpose
date	datetime	Time index
y	numeric	**Target: Hourly temperature**
hour	integer	Hour of day (0-23)
dayofweek	integer	Day of week (0-6)
month	integer	Month (1-12)
quarter	integer	Quarter (1-4)
is_weekend	binary	Weekend indicator
hour_sin	numeric	Cyclical hour encoding
hour_cos	numeric	Cyclical hour encoding
dow_sin	numeric	Cyclical day encoding
dow_cos	numeric	Cyclical day encoding

Final Output: weather_temperature_hourly.csv (8,785 rows × 11 columns)

This provides a standardized dataset that becomes the single input for all models, including baseline, classical, and advanced modeling work. This ensures that all models are trained on and are evaluated with similar data.

4. Methodology

4.1 Baseline Models

Baseline forecasting models are used in this research as simple methods for comparison with other complex forecasting methods. The main goal of using baseline methods for forecasting is not to maximize forecast precision but to set a level of performance that more complex methods should be better. Two types of baseline methods are used: Naïve Forecast, Seasonal Naïve Forecast. The naive forecast assumes that the last observed temperature data point remains constant for all future time periods; that is, the last observed data point for all forecast periods. This model performs surprisingly well for a small forecast horizon for a persistent time series such as temperatures but does poorly for a larger forecast horizon. The seasonal naive model further develops this underlying concept by repeating the most recent observed daily pattern. As hourly temperature time series are shown to be clearly and regularly daily seasonally dependent, temperatures measured in the latest 24-hour cycle are used for a forecast of temperatures for the following 24 hours. Thus, this daily pattern is forecasted for the total forecast period. This model clearly includes a major characteristic of temperature series that is ignored in a simple naive model. As a consequence, a seasonal naive model typically produces a more realistic forecast and performs better than a simple naive forecast model when daily seasonality represents a major feature of series. Taken together, the baseline models provide a clear measure for comparison in determining whether classical and deep learning methods represent a statistically significant improvement.

4.2 Classical Statistical Models

Besides baseline methods, two traditional statistical models used to identify the underlying time series structure of hourly temperature data are Seasonal ARIMA (SARIMA) and Exponential Smoothing (ETS). They have been widely used for forecasting because of their ease of interpretation and robustness. The SARIMA model adds conventional ARIMA by considering both non-seasonal and seasonal components. The model describes short-term temporal relationships using a combination of autoregressive and moving average components and uses seasonal components to formally represent daily periodic behavior. As the data underlying this exercise consists of hourly temperatures measured in a region, a daily seasonal cycle of 24 hours has been used for this model. Differences are introduced for eliminating trends and for obtaining a model that assumes stationarity. The main components of a time series forecasted using the Exponential Smoothing model called Holt-Winters are level, trend, and seasonality. Here, additive trends and additive seasonality are considered because changes in temperatures occur with stable values. ETS methods require less calculation and perform well when used for short-term forecasting. These classical models were chosen because temperature series data often have strong daily seasonality, smooth trends, and strong autocorrelation. Their easy-to-understand model structure makes it convenient to interpret how each component of a time series affects forecasted values.

4.3 Advanced Models

The advanced modelling stage applies Long Short-Term Memory (LSTM) networks to capture nonlinear relationships and longer-range dependencies in hourly temperature data. The final architecture used in this project consists of a single LSTM layer with 32 units, followed by dropout regularization and a dense output layer. This lightweight structure offers a balance between expressive capacity and stable learning, making it well suited to the smooth, seasonal behavior of temperature series.

A univariate setup is adopted so the model learns exclusively from past temperature values, ensuring direct comparability with baseline and classical approaches. Input sequences of 72 hours serve as the lookback window, while predictions are generated for horizons ranging from 24 to 168 hours using an autoregressive strategy. This structure allows the model to exploit short-term fluctuations while retaining awareness of daily cycles.

Training is carried out using the Adam optimizer with a learning rate of 0.0001, batch size of 64, and early stopping to prevent overfitting. These settings produced the most stable convergence during experimentation. Computationally, the shallow LSTM trains quickly and requires far fewer parameters than stacked, bidirectional, or encoder–decoder variants. Although interpretability remains limited compared to classical models, the streamlined architecture provides a clear trade-off between accuracy, efficiency, and reliability.

5. Model Implementation

In order to measure forecasting performance correctly, data were divided in terms of time. This strategy maintained the time scale coherence of weather patterns and avoided the data leakage. Walk-forward validation (so-called rolling-origin strategy) was used to test the models. This was done by fitting the model on a widening window of historical data and extrapolating the following 168 hours (7 days). Each iteration moved the test window forward by 24 hours to simulate an operational forecasting situation in the real world. Specific implementation considerations made to each model architecture are described below:

1. SARIMA: order selection was done with the help of automated step-wise search (Auto-ARIMA) with the aim of reducing the AIC score. This procedure was able to determine the best non-seasonal parameters (p, d, q) and seasonal parameters (P, D, Q, 24) that best modeled daily variations in temperature cycles.

2. ETS: with a state-space approach, the Exponential Smoothing model was used. As the temperature data had comparatively constant seasonality, an additive trend and additive seasonality (AAA) structure was chosen to characterize the information.

3. LSTM: Deep learning involved a Long Short-Term Memory network that was run in PyTorch/Keras. The architecture had a sequential input layer that were trained with multiple epochs in order to discover non-linear dependencies. To make certain of convergence, Adam optimizer and Mean Squared Error loss function were used.

6. Evaluation & Results

The models were tested in seven different forecast horizons between H24 and H168. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used to measure performance and give a complete picture of the quality and level of error.

Comparative Performance The findings show that the classical statistical practices are performing very well compared to the deep learning aspect in this particular dataset.

1. Baseline Models: Naive model was highly competitive, with an RMSE level of 1.705degC being the best at the 24-hour time horizon, and at H96 and H144. On the other hand, the Seasonal Naive method performed poorly as the RMSE values are always greater than 5.0degC which means that a mere 24 hour delay is inadequate in terms of capturing the recent weather changes.

2. Classical Models (ETS and SARIMA): ETS (Error-Trend-Seasonality) turned out to be the most successful. It also got the lowest RMSE levels of most of the horizons (H48, H72, H120, H168), which had a similar error range between 2.0degC and 2.5degC. SARIMA did show a good performance, but it tended to do the same as compared to Seasonal Naive but worse than ETS.

3. Deep Learning (LSTM): In this test the LSTM model showed greater error in comparison with the statistical models and RMSE values were ranging 6.1degC up to 6.8degC. It implies that the model might need to be hyperparameter-tuned or even bigger than ETS to discover the univariate trends such effectively.

Stability and Consistency The comparison of the difference in the error between the different horizons illustrates key information of the robustness of the model.

- Stability: Stability analysis using RMSE variance shows that ETS is the most stable model (Variance, 0.0449), meaning that this model presents the most accurate predictions, regardless of the time period it looks further. Although LSTM was less accurate, it exhibited a reasonable level of stability (Variance: 0.0539) but the least stable was the Seasonal Naive model (Variance: 0.1690).

- Degradation: Generally, there is a decrease in the forecast accuracy with the increase in the horizon. Nevertheless, the ETS model was extremely resilient as the RMSE at H168 (2.009degC) was even better than at H48 (2.583degC) indicating that it represents the weekly seasonality very well.

7. Discussion

The LSTM model underperformed for all forecasting lead-times, with remarkably high values of MAE (6.1-6.6°C) and RMSE (6.3-6.8°C). This result can be due to a lack of data used for training, as well as sub optimal hyper parameters for this specific univariate temperature series. Classical ETS methods performed best for large lead-times (H48, H168), with minimum RMSE of 2.009°C for lead-time H168, while Naive baselines performed best for small lead-times (H24: 1.705°C RMSE). Deep learning captured complex patterns and showed relatively higher degradation for large lead-times than small lead-times (H168/H24 ratio of 1.01), whereas ETS showed remarkably high stability with a variance of 0.0449.

Classical modeling techniques utilized model interpretability through the use of explicit seasonality/trend decomposition, thereby sacrificing a certain amount of forecasting accuracy for transparency essential in energy/HVAC applications. Occasionally, the baseline models outperformed advanced methods for a longer-term forecasting period, which can be attributed to the characteristics of the stations being constant and the law of diminishing returns due to the increased complexity of LSTMs. These models are appropriate for Short-term forecasting, which falls into daily activities (Naive model), while medium-term forecasting requires use of ETS models.

8. Limitations

Data limitation conditions involve a univariate approach that omits humidity and pressure features, which can limit model expressiveness but facilitate comparative analyses among different models. The resampling approach from high-frequency data (10-minute) to hourly data can suppress micro-variations, hence causing aggregation bias in cases of data missingness. SARIMA and ETS models are based on the assumptions of stationarity and linearity, hence performing sub-optimally on scenarios with non-stationary shifts; in addition, LSTM models involve heavy hyper-tuning, considering that a small data set of about 8,000 hourly values was used.

Computational demands limit grid search and ensemble analyses of LSTMs, due to impractical training times for a real-time model. The model evaluation, fixed in a one-year perspective, ignored multi-year seasonality and climatic variations, hence limited generalizability in a 2020-2021 environment.

9. Conclusion

In this case, ETS performs best for 1-7 day temperature forecasts, in terms of a balance of accuracy, in particular for H48+ values, with stability, speed, and interpretation. The naive models are sufficient for accuracy in very short-term forecasts, whereas the rate of performance as well as inaccuracy of LSTMs are slower without multivariate models and bigger data.

To move ahead, it would be important that future studies include exogenous variables like wind and radiation, use multi-step validation, and develop hybrid models that combine the strengths of ETS models and LSTM models. Moreover, it would also be useful to use ensemble averaging with different forecast periods and apply transfer learning with regional weather patterns.

10. Resources & Code Repository

Dataset:

https://www.kaggle.com/datasets/alistairking/weather-long-term-time-series-forecasting

- Continuous hourly air temperature series (2020-01-01 to 2021-01-01)
- Target: y (hourly average temperature °C)
- Features: Calendar (hour, dayofweek, month, quarter, is_weekend) + cyclical encodings (hour_sin, hour_cos, dow_sin, dow_cos)

GitHub Repository:

https://github.com/aejae-da/weather-forecasting-project-gha.git

Contains:

- Raw 10-minute weather data
- Processed hourly dataset
- All Jupyter notebooks (01_data_preprocessing, 02_baseline_classical, 03_advanced_LSTM, 04_evaluation, 05_final_report)
- Model results & forecasts CSVs
- Complete reproducible pipeline